

Systems of Equations

Jacobi Method:

Solve $3u + v = 5, u + 2v = 5$.

$$u = \frac{5 - v}{3} \quad v = \frac{5 - u}{2}.$$

initial guess $(u_0, v_0) = (0, 0)$.

The two equations are iterated:

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} u_1 \\ v_1 \end{bmatrix} = \begin{bmatrix} \frac{5-v_0}{3} \\ \frac{5-u_0}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-0}{3} \\ \frac{5-0}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{5}{2} \end{bmatrix}$$

$$\begin{bmatrix} u_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{5-v_1}{3} \\ \frac{5-u_1}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-5/2}{3} \\ \frac{5-5/3}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{6} \\ \frac{5}{3} \end{bmatrix}$$

$$\begin{bmatrix} u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} \frac{5-5/3}{3} \\ \frac{5-5/6}{2} \end{bmatrix} = \begin{bmatrix} \frac{10}{9} \\ \frac{25}{12} \end{bmatrix}.$$

Further steps of Jacobi show convergence toward the solution, which is $[1, 2]$.

Equations in the reverse order

$$u + 2v = 5, 3u + v = 5.$$

$$u = 5 - 2v$$

$$v = 5 - 3u.$$

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} u_1 \\ v_1 \end{bmatrix} = \begin{bmatrix} 5 - 2v_0 \\ 5 - 3u_0 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} u_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} 5 - 2v_1 \\ 5 - 3u_1 \end{bmatrix} = \begin{bmatrix} -5 \\ -10 \end{bmatrix}$$

$$\begin{bmatrix} u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 5 - 2(-10) \\ 5 - 3(-5) \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \end{bmatrix}.$$

In this case the Jacobi Method fails, as the iteration diverges.

Theorem:

If the $n \times n$ matrix A is strictly diagonally dominant, then (1) A is a nonsingular matrix, and (2) for every vector b and every starting guess, the Jacobi Method applied to $Ax = b$ converges to the (unique) solution. 

The $n \times n$ matrix $A = (a_{ij})$ is **strictly diagonally dominant** if, for each $1 \leq i \leq n$, $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$.

Example: $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -5 & 2 \\ 1 & 6 & 8 \end{bmatrix}$

Solve:

$$3u + v = 5, u + 2v = 5.$$

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \quad (\text{Diagonally dominant})$$

$$Ax = b$$

$$u + 2v = 5, 3u + v = 5.$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \quad (\text{Not diagonally dominant})$$

The **spectral radius** $\rho(A)$ of a square matrix A is the maximum magnitude of its eigenvalues.

Fact:

If the $n \times n$ matrix A has spectral radius $\rho(A) < 1$, and b is arbitrary, then, for any vector x_0 , the iteration $x_{k+1} = Ax_k + b$ converges.

$$x_1 = Ax_0 + b, \quad x_0 = \sum_{i=1}^n a_i v_i \quad (v_i \text{'s are the eigenvectors of } A)$$

$$= \sum_{i=1}^n a_i \lambda_i v_i + b, \quad |\lambda_i| < 1$$

Because $|\lambda_i| < 1$

$$x_2 = Ax_1 + b = A^2 x_0 + Ab + b$$

$$x_n = \sum_{i=1}^n \lambda_i^n a_i v_i + \dots \text{ converges}$$

$$= \sum_{i=1}^n \lambda_i^2 a_i v_i + \sum_{i=1}^n c_i \lambda_i v_i + b, \quad b = \sum_{i=1}^n c_i v_i$$

Proof of the theorem:

$$Ax = b \text{ --(1)}$$

$$A = L + U + D$$

(**L** and **U** are lower and upper triangular matrices whose diagonal elements are zero.

D is a diagonal matrix.)

$$(L + U + D)x = b$$

$$Dx = b - (L + U)x$$

$$x = D^{-1}(b - (L + U)x)$$

$$x_{k+1} = D^{-1}b - D^{-1}(L + U)x_k$$

$$\text{Let } R = L + U$$

To check $\rho(D^{-1}R) < 1$, let λ be an eigenvalue of $D^{-1}R$ with corresponding eigenvector v .

Choose this v so that $\|v\|_\infty = 1$, ($\max\{v_k: 1 \leq k \leq n\}$) so that for some $1 \leq m \leq n$, the component $v_m = 1$

$$D^{-1}Rv = \lambda v, \text{ or } Rv = \lambda Dv.$$

Since $r_{mm} = 0$, absolute values of the m th component

$$\begin{aligned} |r_{m1}v_1 + r_{m2}v_2 + \cdots + r_{m,m-1}v_{m-1} + r_{m,m+1}v_{m+1} + \cdots + r_{mn}v_n| \\ = |\lambda d_{mm}v_m| = |\lambda||d_{mm}|. \end{aligned}$$

Since all $|v_i| \leq 1$, the left-hand side is at most $\sum_{j \neq m} |r_{mj}| < |d_{mm}|$,

$$\Rightarrow |\lambda||d_{mm}| < |d_{mm}|,$$

$$|\lambda| < 1.$$

$$\rho(D^{-1}R) < 1$$

Finally, since $Ax = b$ has a solution for arbitrary b , A is a nonsingular matrix.

Gauss-Seidel Method:

Solve $3u + v = 5, u + 2v = 5$.

$$u = \frac{5 - v}{3} \quad v = \frac{5 - u}{2}.$$

initial guess $(u_0, v_0) = (0, 0)$.

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} u_1 \\ v_1 \end{bmatrix} = \begin{bmatrix} \frac{5-v_0}{3} \\ \frac{5-u_1}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-0}{3} \\ \frac{5-5/3}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{5}{3} \end{bmatrix}$$

$$\begin{bmatrix} u_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{5-v_1}{3} \\ \frac{5-u_2}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-5/3}{3} \\ \frac{5-10/9}{2} \end{bmatrix} = \begin{bmatrix} \frac{10}{9} \\ \frac{35}{18} \end{bmatrix}$$

$$\begin{bmatrix} u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} \frac{5-v_2}{3} \\ \frac{5-u_3}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-35/18}{3} \\ \frac{5-55/54}{2} \end{bmatrix} = \begin{bmatrix} \frac{55}{54} \\ \frac{215}{108} \end{bmatrix}$$

Nonlinear Systems of Equations (Newton's Method):

The one-variable Newton's Method

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Taylor expansion of a multivariable function

$$f(u, v, w) = f(u_0, v_0, w_0) + \frac{\partial f}{\partial u} \Delta u + \frac{\partial f}{\partial v} \Delta v + \frac{\partial f}{\partial w} \Delta w$$

Multivariate Newton's Method

$$f_1(u, v, w) = 0$$

$$f_2(u, v, w) = 0$$

$$f_3(u, v, w) = 0$$

$$F(u, v, w) = (f_1, f_2, f_3) \quad (\text{vector-valued function})$$

$$F(x) = 0, \text{ where } x = (u, v, w).$$

$$F(x) = F(x_0) + DF(x_0) \cdot (x - x_0) \quad \text{where } x_0 = (u_0, v_0, w_0)$$

$$DF(x) = \begin{bmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial w} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial w} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial w} \end{bmatrix}.$$

$$F(x) = F(x_0) + DF(x_0) \cdot (x - x_0)$$

let $x = r$ be the root, and let x_0 be the current guess. Then

$$0 = F(r) \approx F(x_0) + DF(x_0) \cdot (r - x_0),$$

$$-DF(x_0)^{-1}F(x_0) \approx r - x_0.$$

$$x_{k+1} = x_k - (DF(x_k))^{-1}F(x_k) \quad \textbf{for } k = 0, 1, 2, \dots$$

$$DF(x) = \begin{bmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial w} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial w} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial w} \end{bmatrix}.$$